

Viral Hepatitis

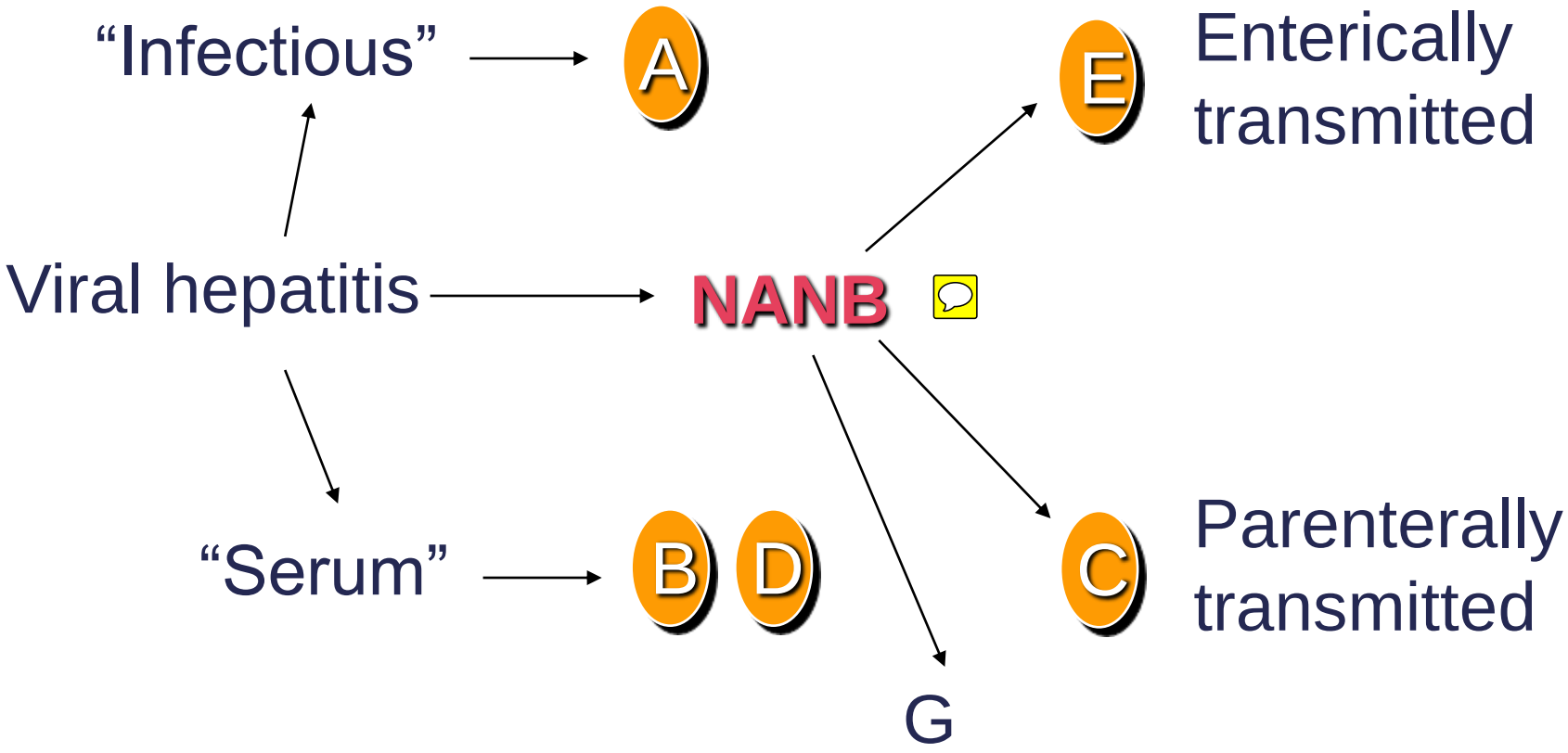
What is Viral Hepatitis?

- Viral hepatitis is a systemic disease with primary inflammation of the liver by any one of a heterogeneous group of hepatotropic viruses.
- The most common causes of viral hepatitis are the five unrelated hepatotropic viruses Hepatitis A, Hepatitis B, Hepatitis C, Hepatitis D, and Hepatitis E.
- In addition to the nominal hepatitis viruses, other viruses that can also cause liver inflammation include Herpes simplex, Cytomegalovirus, Epstein–Barr virus, or Yellow fever.

❖ Clinical Terms

- Hepatitis: inflammation of liver;
- Acute Viral Hepatitis: symptoms last less than 6 months
- Acute Hepatic Failure:is the appearance of severe complications rapidly after the first signs of liver disease (such as jaundice), and indicates that the liver has sustained severe damage (loss of function of 80-90% of liver cells).Massive hepatic necrosis with impaired consciousness within 8 weeks of onset of illness.
- Chronic Hepatitis: Inflammation of liver for at least 6 months
- Cirrhosis: Replacement of liver tissue→ fibrosis(scar tissue).These changes lead to loss of liver function.
- Fulminant Hepatitis: severe impairment of hepatic functions or severe necrosis of hepatocytes in the absence of preexisting liver disease.

Viral Hepatitis- Historical Perspective



❖ Types Of Viral Hepatitis

	Viral Hepatitis A	Viral Hepatitis B	Viral Hepatitis C	Viral Hepatitis D	Viral Hepatitis E
Agent	Hepatitis A virus (HAV); ssRNA; No envelope	Hepatitis B virus (HBV); dsDNA; envelope	Hepatitis C virus (HCV); ssRNA; envelope	Hepatitis D virus (HDV); ssRNA; envelope from HBV	Hepatitis E virus (HEV); ssRNA; no envelope
Route of Transmission	Fecal-oral	Parenteral, Vertical, Sexual.	Parenteral	Parenteral	Fecal-oral
Age affected	Children	Any age	Adults	Any age	Young adults
Carrier state	Nil	Common	Present	Nil (only with HBV)	Nil
Incubation period	10-50 days (avg. 25-30)	50-180 days (avg. 60-90)	40-120 days	2-12 weeks	2-9 weeks
Chronic infection	No	Yes	Yes	Yes	No
Specific Prophylaxis	Ig and Vaccine	Ig and Vaccine	Nil	HBV vaccine	Nil

Hepatitis A

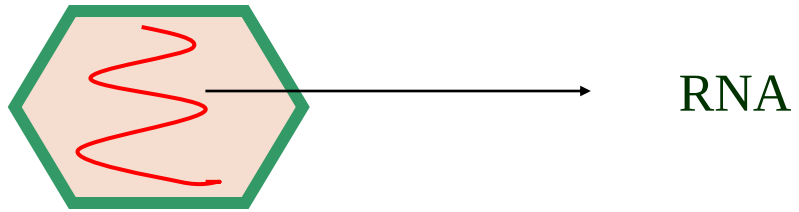
❖Hepatitis A

- ✓ Hepatitis A (formerly known as “infectious” hepatitis or epidemic jaundice) is an acute infectious disease caused by Hepatitis A virus (HAV).
- ✓ The disease is heralded by non-specific symptoms such as fever, chills, headache, fatigue, generalized weakness and aches and pains, followed by anorexia, nausea, vomiting, dark urine and jaundice.
- ✓ The disease is benign with complete recovery in several weeks.

❖ Epidemiological determinants

➢Agent factors

a) AGENT: The causative agent, the hepatitis A virus, is an enterovirus of the Picornaviridae family. It multiplies only in hepatocytes.
b) RESISTANCE: The virus is fairly resistant to heat and chemicals.
-Withstands heating to 60^0 C for 1 hr. and is not affected by chlorine in doses usually employed for chlorination.
-Formalin is stated to be an effective disinfectant.
-The virus is inactivated by ultraviolet rays and by boiling for 5 minutes or autoclaving.



Naked RNA virus

c) RESERVOIR OF INFECTION: The human cases are the only reservoir of infection.

d) PERIOD OF INFECTIVITY : The risk of transmitting HAV is greatest from 2 weeks before to 1 week after the onset of jaundice.

e) INFECTIVE MATERIAL : Mainly man's faeces.

f) VIRUS EXCRETION: HAV is excreted in the faeces for about 2 weeks before onset of jaundice and for up to 2 weeks thereafter.

➢Host factors

a) AGE: Infection with HAV is more frequent among children than in adults. However, people from all ages may be infected if susceptible.

b) SEX: Both sexes are equally susceptible.

c) IMMUNITY: Immunity after attack probably lasts for life.

➢Environmental factors

- Cases may occur throughout the year.
- In India the disease tends to be associated with periods of heavy rainfall.

❖Incubation period

10-50 days (usually 25 to 30 days).

❖Mode of Transmission

a) *FAECAL-ORAL ROUTE:* Major route of transmission.
   -By contaminated water, food or milk.

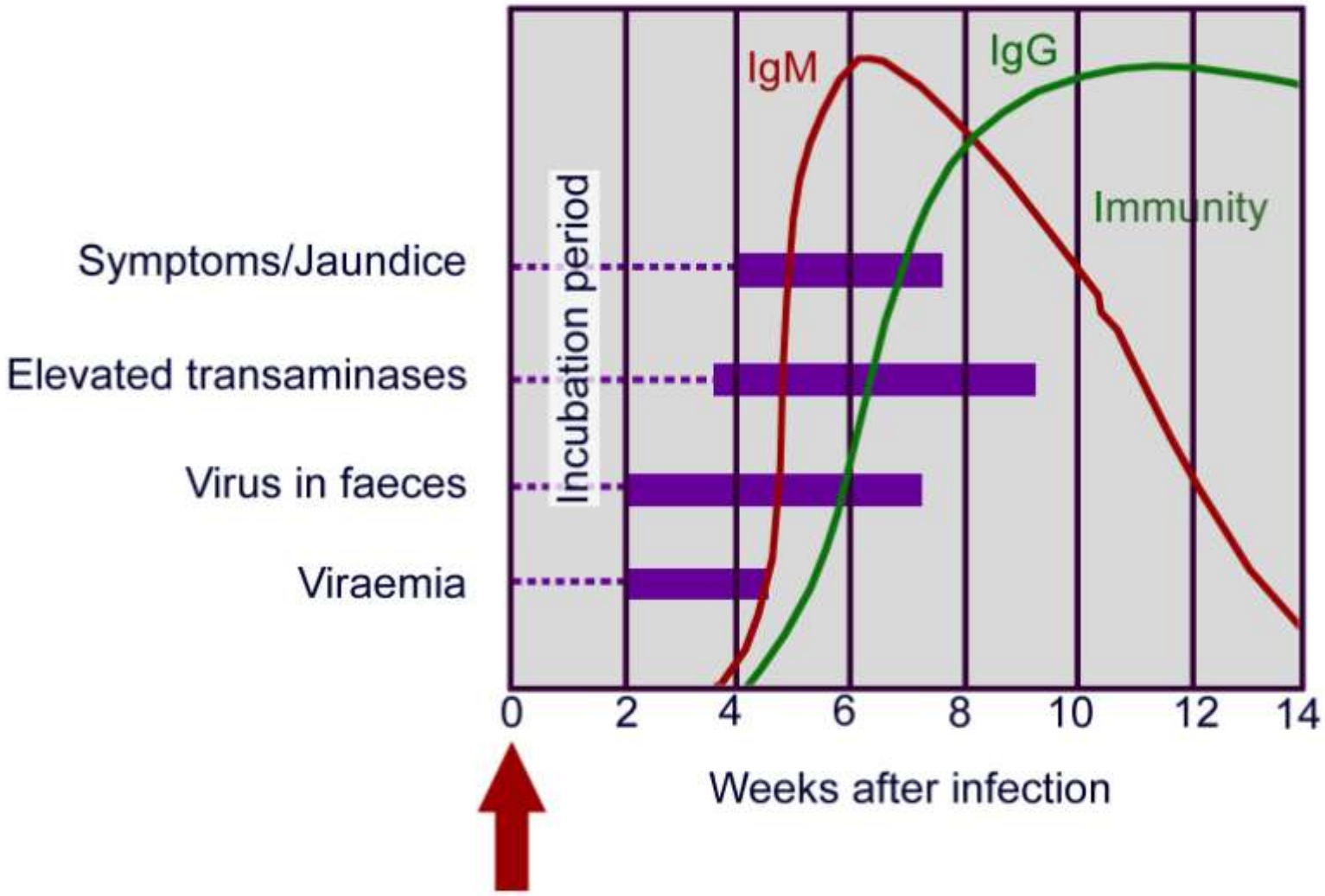
b) *PARENTERAL ROUTE (Rarely):*
   -By blood and blood products or by skin penetration through contaminated needles.

c) *SEXUAL TRANSMISSION:*
   -May occur mainly among homosexual men because of oral-anal contact.

❖Diagnosis

1. Demonstration of Virus in feces, blood, bile:
   By: Immunoelectron microscopy
2. Virus Isolation:
3. Detection of Antibody :By ELISA
4. Biochemical tests:
   i) Alanine aminotransferase (ALT)
   ii) Bilirubin
   iii) Protein
5. Molecular Diagnosis : RT PCR of feces

Hepatitis A time course of infection



❖Prevention:-

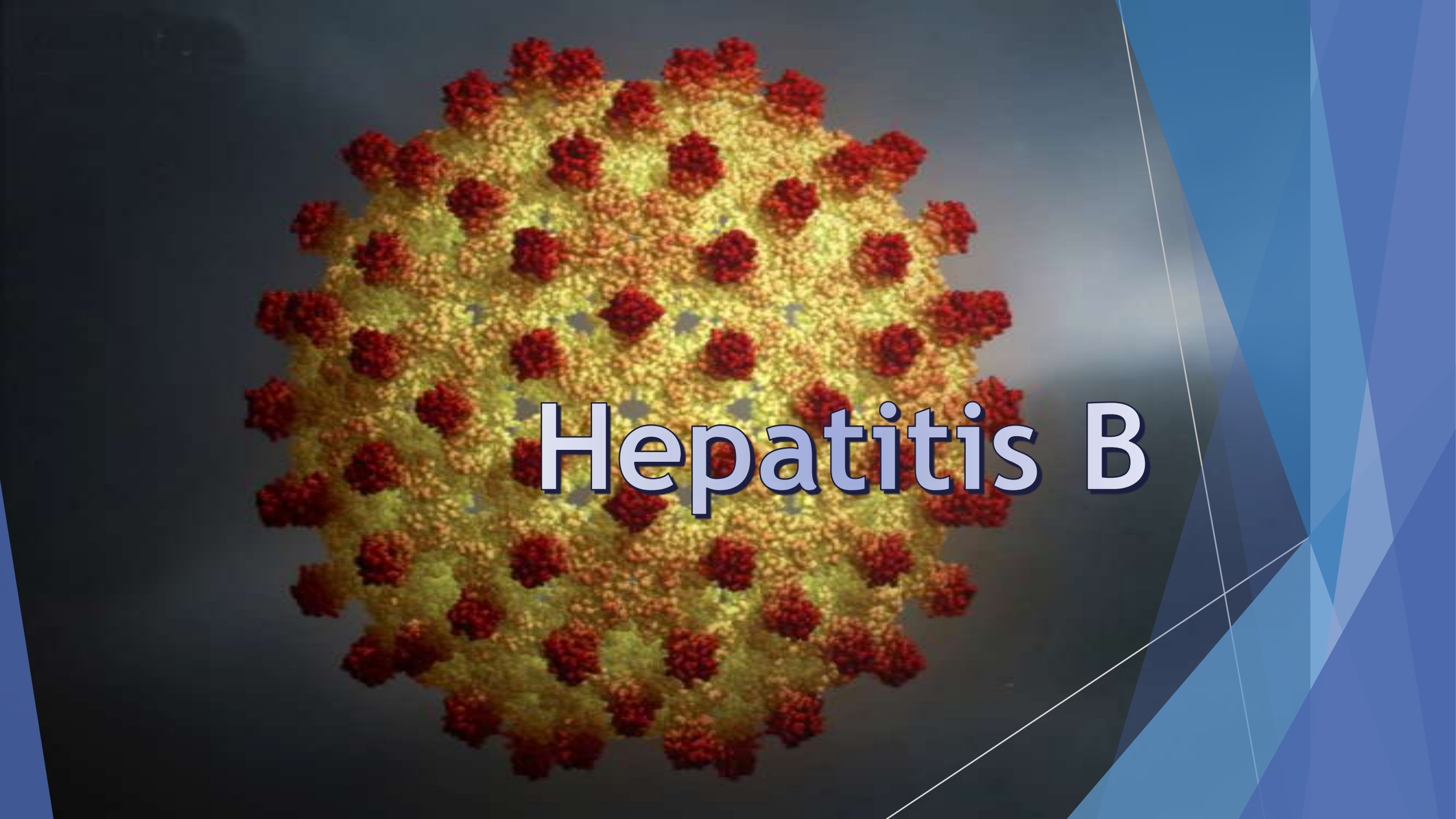
-hygienic measures and sanitation

-passive immunization(Human Immunoglobulin
Gamma globulin given before exposure to virus or early during the incubation period, will prevent or attenuate a clinical illness.

-active immunization
Several inactivated or live attenuated vaccines against hepatitis A have been developed.

❖Treatment:

-nospecific, dietary food and long rest


Hepatitis B

❖Hepatitis B

- ✓Hepatitis B (formerly known as “serum” hepatitis) is an acute systemic infection with major pathology in the liver, caused by hepatitis B virus.
- ✓Transmitted by the Parenteral route.
- ✓The acute illness causes liver inflammation, vomiting, jaundice, and, rarely, death. Chronic hepatitis B may eventually cause cirrhosis and liver cancer.
- ✓Hepatitis B is endemic throughout the world, especially in tropical & developing countries.

❖Epidemiology Determinants

➢Agent factor

a) AGENT: Hepatitis B Virus (HBV)

-It is a complex, 42 nm double-shelled DNA virus originally known as "Dane Particle".
-It replicates in liver cell.

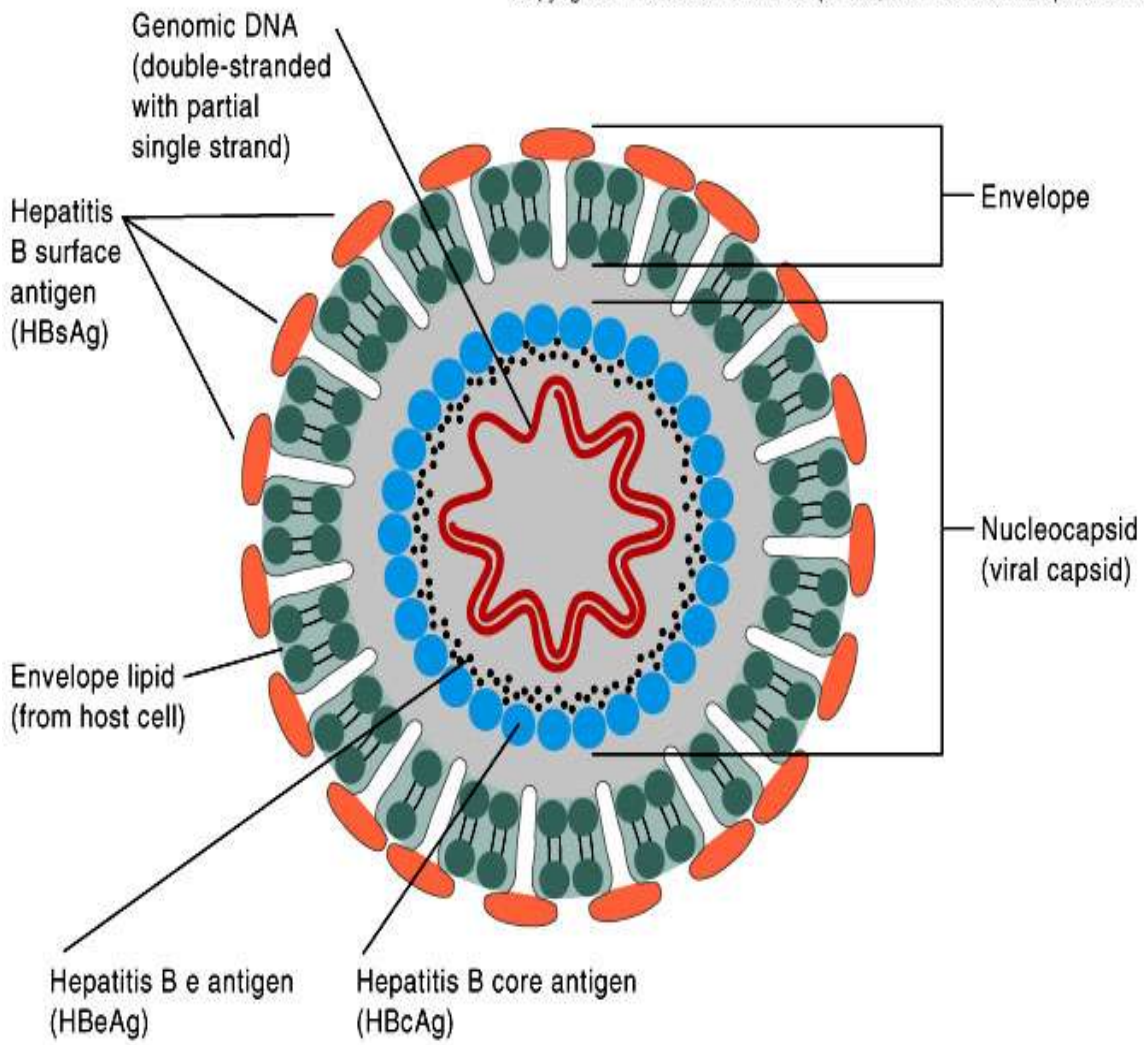
HBV occurs in 3 morphology form in serum:

i. Small spherical particles with an average Diameter of 22nm.
ii. Filamentous or Tubules of varying length & of 22 nm diameter.
iii. Dane particle.

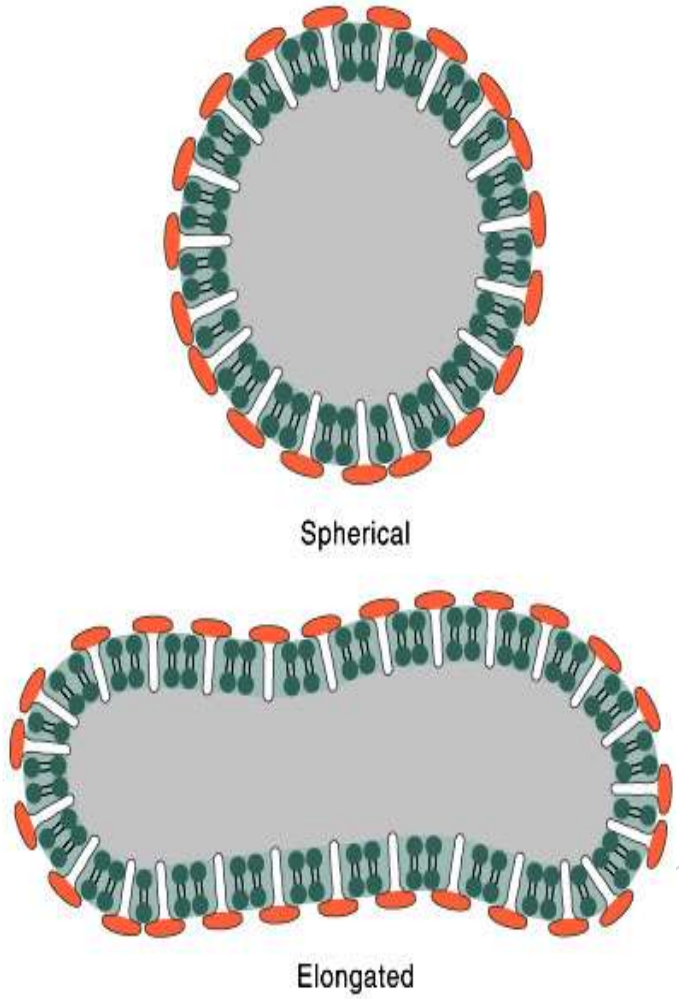
Out of 3 morphology forms, only the Dane particle is considered infectious, other circulating morphology forms are not infectious.

❖HBV : Structure

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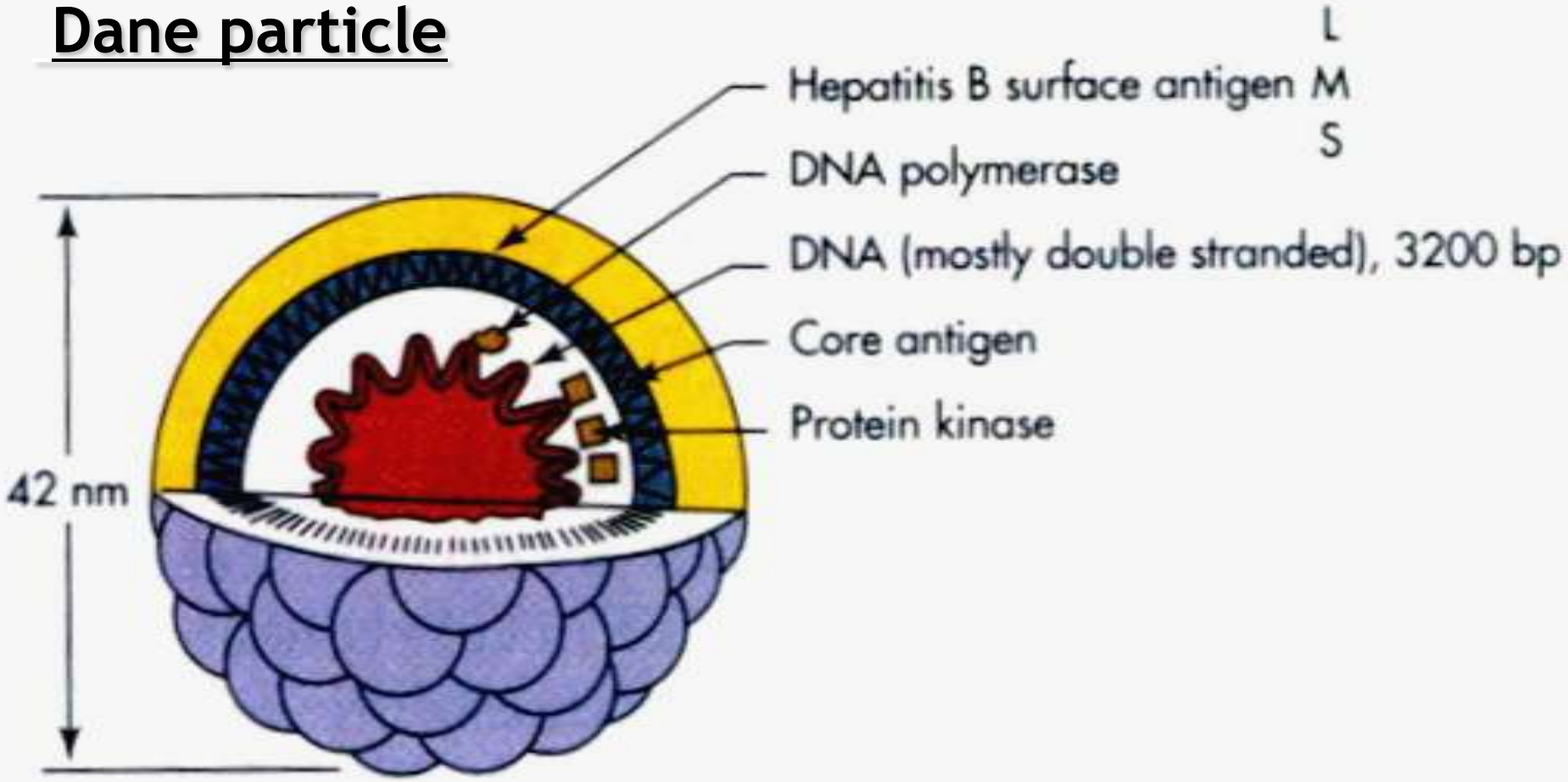


(a) Complete infectious virion



(b) Viral envelope particles containing HBsAg

Dane particle



b) RESERVOIR OF INFECTION:

-Men is the only reservoir of infection which can be spread either from carriers or from cases.

c) Infective material:

-Contaminated blood is the main source,

-Virus has been found in body secretion such as saliva, vaginal secretion & Semen in infected material.

d) Resistance:

-Readily destroyed by sodium hypochlorite, as is by heat sterilization in an autoclave for 30-60 min.

➢Host factor

a) AGE:

-***Acute hepatitis B***

90% resolve by themselves; <1% develop fulminant hepatic failure.

-occurs in approx.:

Perinatal	-1%
Childhood	-10%(1-5 yr. age)
Late infection	-30%(>5 yr. age)

-***Chronic hepatitis B***

2-10% progress to chronic state.

-occur in approx.

Perinatal	-95%
Childhood	-80%
After 5 yr. of age	-5-10%

b) High Risk Group:

- ✓ People from endemic regions
- ✓ Babies of mothers with chronic HBV
- ✓ Intravenous drug abusers
- ✓ People with multiple sex partners
- ✓ Hemophiliacs and other patients requiting blood and blood product treatments
- ✓ Health care personnel who have contact with blood
- ✓ Patients who are immunocompromised.

c) Humoral and cellular response:

-HBV has 3 distinct antigen:

i. HBsAg, also known as “Australian antigen,
ii. HBcAg antigen (core antigen)
iii. HBeAg envelope antigen

They stimulate production of corresponding antibody.

❖Incubation Period

45-180 days (usually 60-90 days)

❖Mode of Transmission

- ✓ Parenteral- IV drug abusers, health workers are at increased risk.
- ✓ Sexual- sex workers and homosexuals are particular at risk.
- ✓ Perinatal (**Vertical**) - mother (HBeAg+) →infant. Mothers who are HBeAg positive are much more likely to transmit to their offspring than those who are not. Perinatal transmission is the main means of transmission in high prevalence populations.

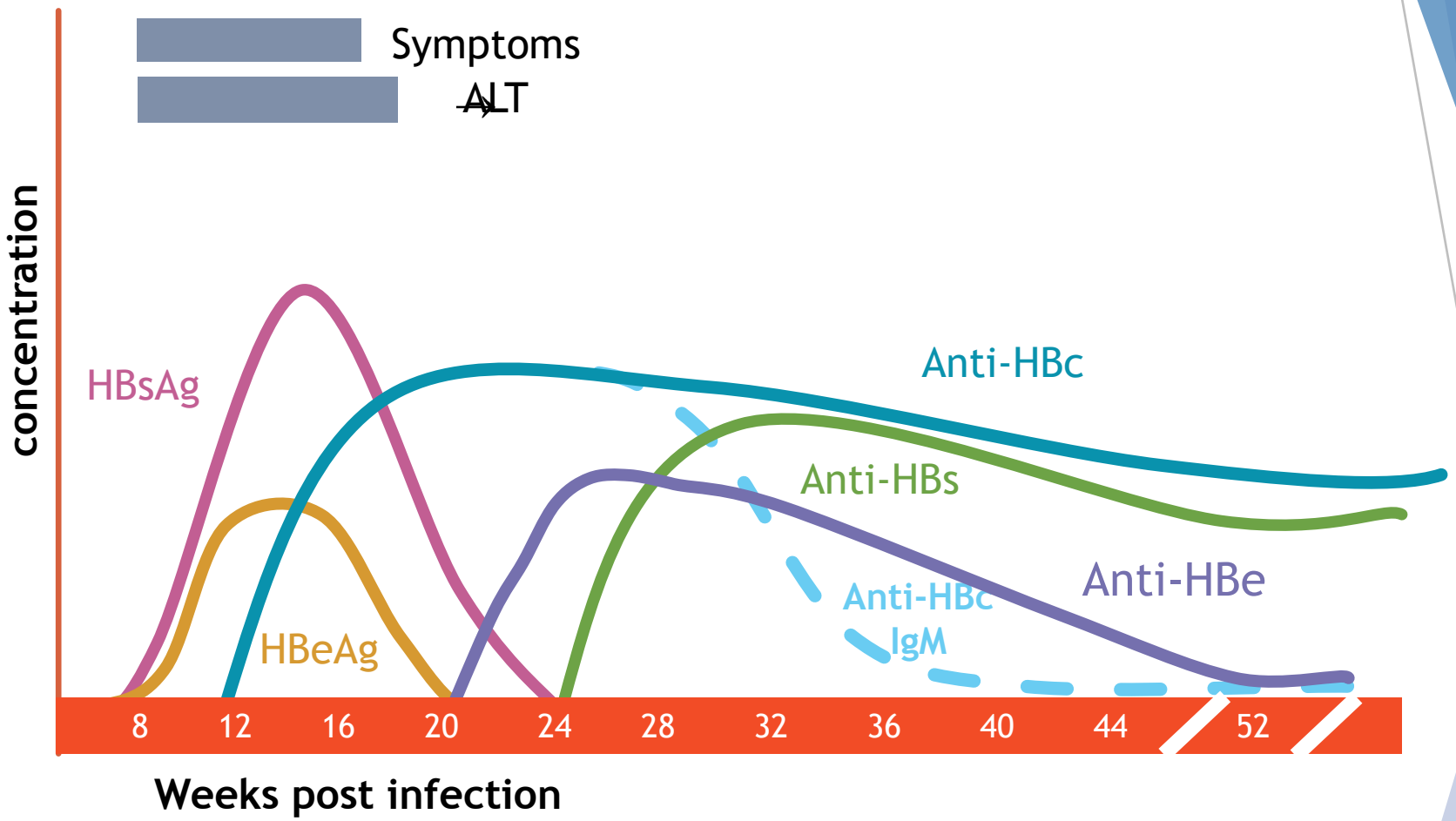
❖Diagnosis

- ✓ Serology
- ✓ Liver Chemistry tests
  - AST, ALT, ALP, and total Bilirubin
- ✓ Histology--Immunoperoxidase staining
- ✓ HBV Viral DNA--Most accurate marker of viral DNA and detected by PCR
- ✓ Liver Biopsy--to determine grade(Inflammation) and stage(Fibrosis) in chronic Hepatitis

❖Serologic Events

1) HBsAg :- It is the first marker to appear in blood after infection.
2) Anti-HBs(HBsAb) :-Disappearance of HBsAg and the appearance of anti-HBs signals recovery from HBV infection, non-infectivity.
3) Anti-HBc :- IgM anti-HBc appears shortly after HBsAg is detect (HBcAg alone dose not appear in serum)
   IgM-HBc may also or can persist for 3-6 months or longer.
   IgG-HBc also appear during acute hepatitis B but persist indefinitely.
4) HBeAg :-

   → HBeAg appear in blood concurrently with HBsAg, or soon afterwards.
   → HBeAg is a soluble protein found only in HBeAg positive serum.
   → HBeAg indicate viral replication and infectivity.
   → Persistence of HBeAg in serum beyond 3 month indicate an increased like hood of chronic hepatitis B.


Symptoms
ALT
concentration
HBsAg
HBeAg
Anti-HBc
Anti-HBs
Anti-HBe
Anti-HBc IgM
8
12
16
20
24
28
32
36
40
44
52
Weeks post infection

❖Interpretation of common serological patterns in HBV infection

Virus/Antibody markers					Interpretation
HBsAg	HBeAg	Anti-HBc	Anti-HBs	Anti-HBe	
+	+	IgM	-	-	Acute HBV infection; highly infectious
+	+	IgG	-	-	Late/Chronic HBV infection or carrier state; highly infectivity
+	-	IgG	-	+/-	Late/Chronic HBV infection or carrier state; low infectivity
-	+/-	IgM	-	+/-	Seen rarely in early acute HBV infection; infectious
-	-	IgG	+/-	+/-	Remote HBV infection; infectivity nil or very low
-	-	-	+	-	Immunity following HBV vaccine

❖Prevention

- **Vaccination**

  - highly effective recombinant vaccines

- **Hepatitis B Immunoglobulin** (HBIG)

  -exposed within 48 hours of the incident/ neonates whose mothers are HBsAg and HBeAg positive.

- **Other measures**

  -screening of blood donors, blood and body fluid precautions.

❖National Immunization Schedule

Vaccine	When to Give	Dose	Route	Site
Hepatitis B	At birth or as soon as possible with in 24 hours.	0.5 ml	IM	Antero lateral side of mid thigh
Hepatitis B 1,2,3	At 6, 10, 14 weeks	0.5 ml	IM	Antero lateral side of mid thigh

❖Treatment

- ✓ Interferon Alfa (Intron A)
  Response rate is 30 to 40%.
- ✓ Lamivudine (Epivir HBV)
  (relapse ,drug resistance)
- ✓ Adefovir dipivoxil (Hepsera)

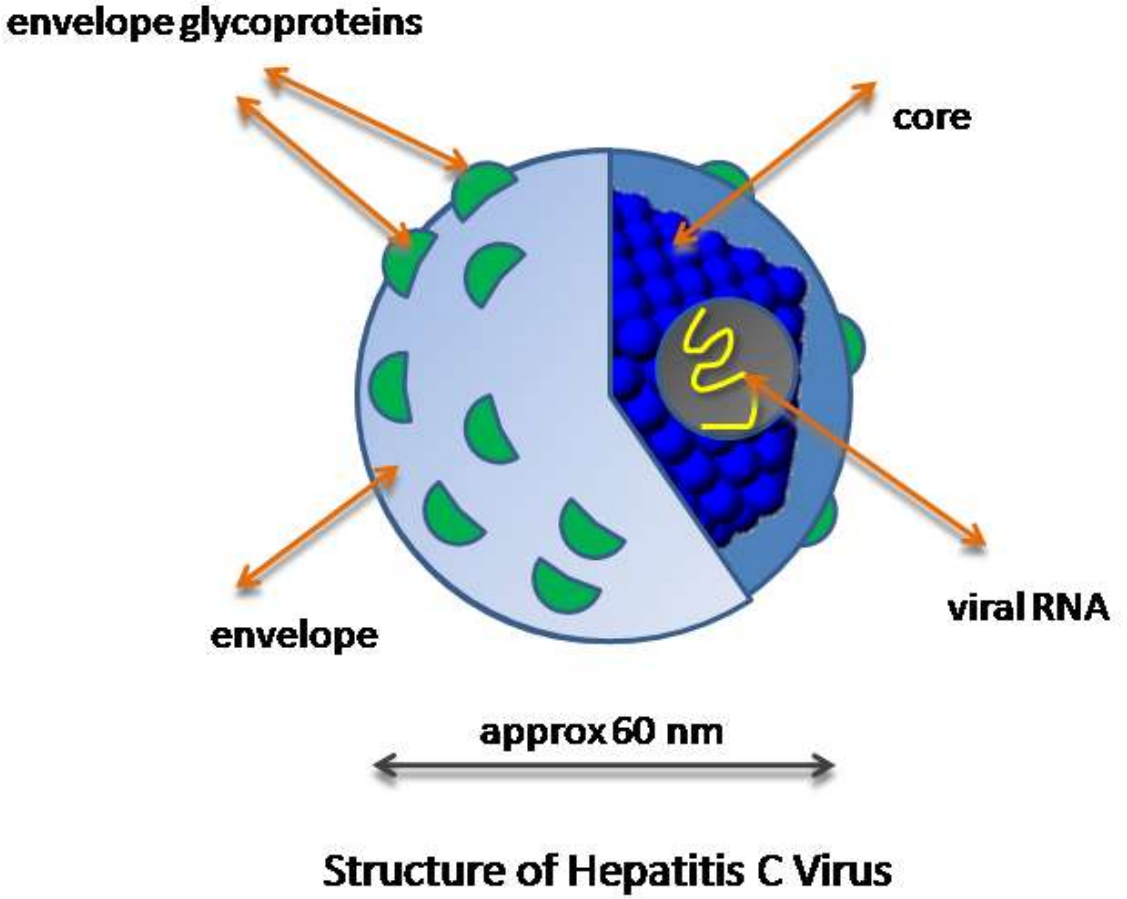
Hepatitis C

❖Hepatitis C

- ✓ **Hepatitis C** is an infectious disease affecting primarily the liver, caused by the hepatitis C virus (HCV).
- ✓ The infection is often asymptomatic, but chronic infection can lead to scarring of the liver and ultimately to cirrhosis, which is generally apparent after many years.
- ✓ It is estimated that 150–200 million people, or ~3% of the world's population, are living with chronic hepatitis C.
- ✓ HCV infection is prevalent in India too, with an estimated 12.5 million cases.
- ✓ Overt jaundice is seen in about 5 % of patients only.
- ✓ The important part in type C hepatitis is the chronic illness.
- ✓ About 50 to 80 % of patients progress to chronic hepatitis.

❖Hepatitis C Virus

- ✓ HCV is a 50-60 nm virus with a linear, single stranded RNA genome, enclosed with in a core and surrounded by an envelope, carrying glycoprotein spikes.
- ✓ It is a member of the *Hepacivirus* genus in the family *Flaviviridae*.
- ✓ The half life of the virus particles in the serum is around 3 hours and may be as short as 45 minutes.
- ✓ In addition to replicating in the liver the virus can multiply in lymphocytes.



Structure of Hepatitis C Virus

❖Incubation Period

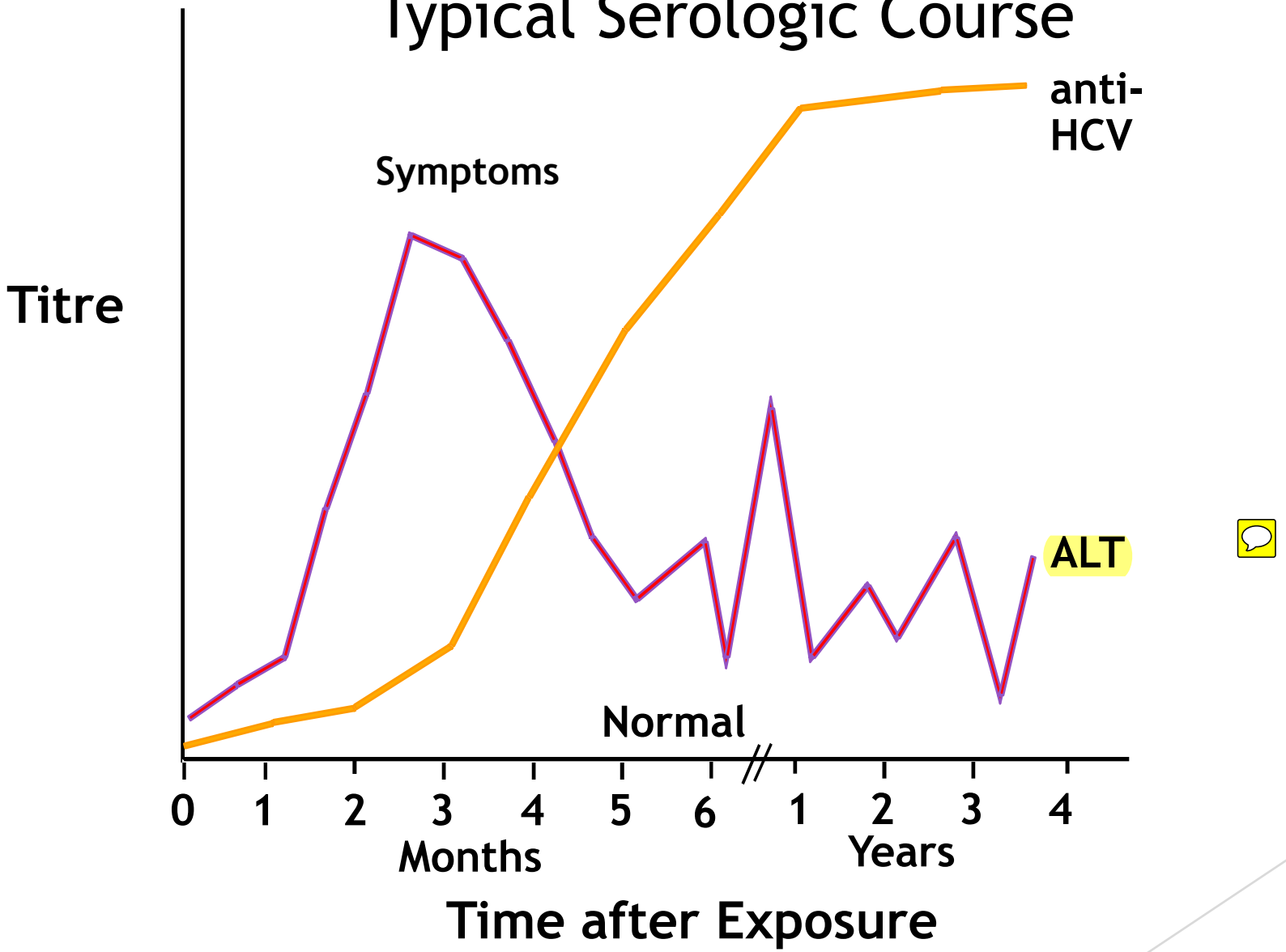
40-120 days

❖Mode of Transmission

- ✓ Intravenous Drug Use
- ✓ Healthcare Exposure: Blood Transfusion, transfusion of Blood products, Organ Transplant without HCV screening carry significant risk of infection.
- ✓ Hemodialysis
- ✓ Accidental injuries with needles/sharps
- ✓ Sexual/household exposure to anti-HCV-positive contact
- ✓ Multiple sex partners
- ✓ Vertical Transmission: Vertical transmission of hepatitis C from an infected mother to her child

Hepatitis C Virus Infection

Typical Serologic Course



❖Diagnosis

- HCV antibody – ELISA used to diagnose hepatitis C infection. Not useful in the acute phase as it takes at least 4 weeks after infection before antibody appears.

- HCV-RNA - various techniques are available e.g. PCR and branched DNA. May be used to diagnose HCV infection in the acute phase. However, its main use is in monitoring the response to antiviral therapy.

- HCV-antigen - an EIA for HCV antigen is available. It is used in the same capacity as HCV-RNA tests but is much easier to carry out.

❖Prevention

- ✓ Only General Prophylaxis, such as blood, tissue, organ screening, is possible.
- ✓ No specific active or passive immunizing agent is available.

❖Treatment

Interferon - may be considered for patients with chronic active hepatitis. The response rate is around 50% but 50% of responders will relapse upon withdrawal of treatment.

Ribavirin - there is less experience with ribavirin than interferon. However, recent studies suggest that a combination of interferon and ribavirin is more effective than interferon alone.

Hepatitis D

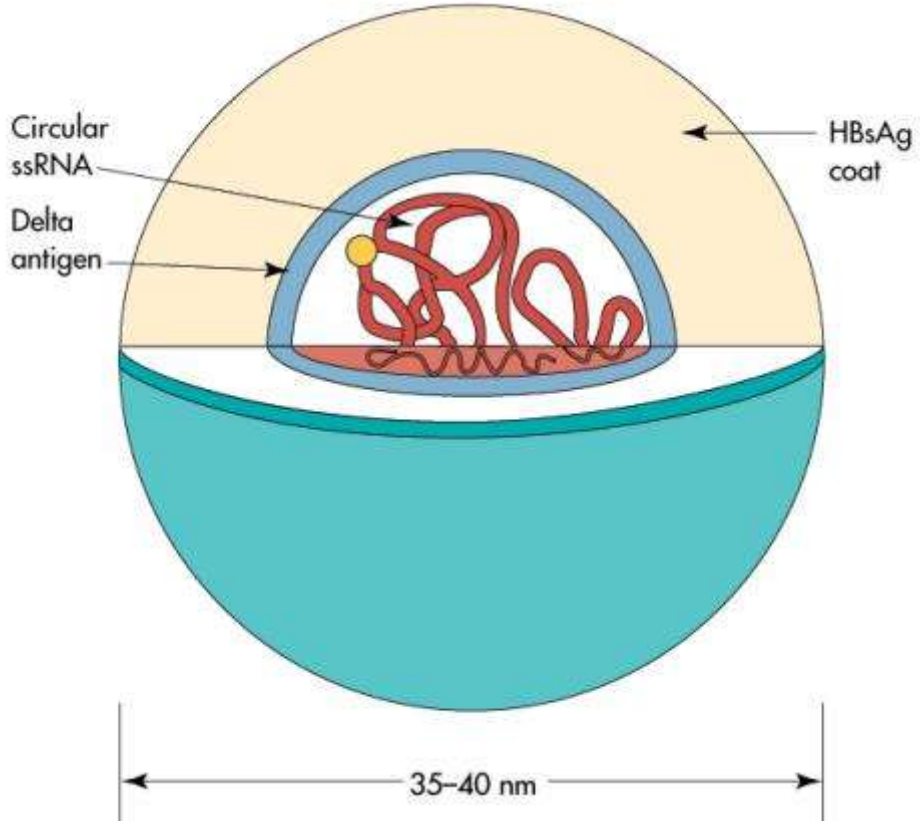
❖Hepatitis D

- ✓ **Hepatitis D**, also referred to as **hepatitis D virus** (**HDV**) and classified as *Hepatitis delta virus*, is a disease caused by a small circular enveloped RNA virus.
- ✓ HDV is considered to be a subviral satellite because it can propagate only in the presence of the hepatitis B virus (HBV).

❖Hepatitis D virus

VIRION: spherical, 36-38 nm particle with an outer coat composed of the HBsAg surrounding ssRNA genome.

Satellite virus : replicates only in the presence of HBV



❖Incubation Period

2-12 weeks

❖Mode of Transmission

The primary route of Transmission are believed to be similar to those of HBV, though HDV does not appear to be sexually transmitted disease.

❖Clinical Features

- ✓ Infection is dependent on HBV replication, as HBV provides an HBsAg envelop for HDV.
- ✓ Two types of infection are recognisesd, coinfection and superinfection.

In Coinfection, delta and HBV are transmitted together at the same time.

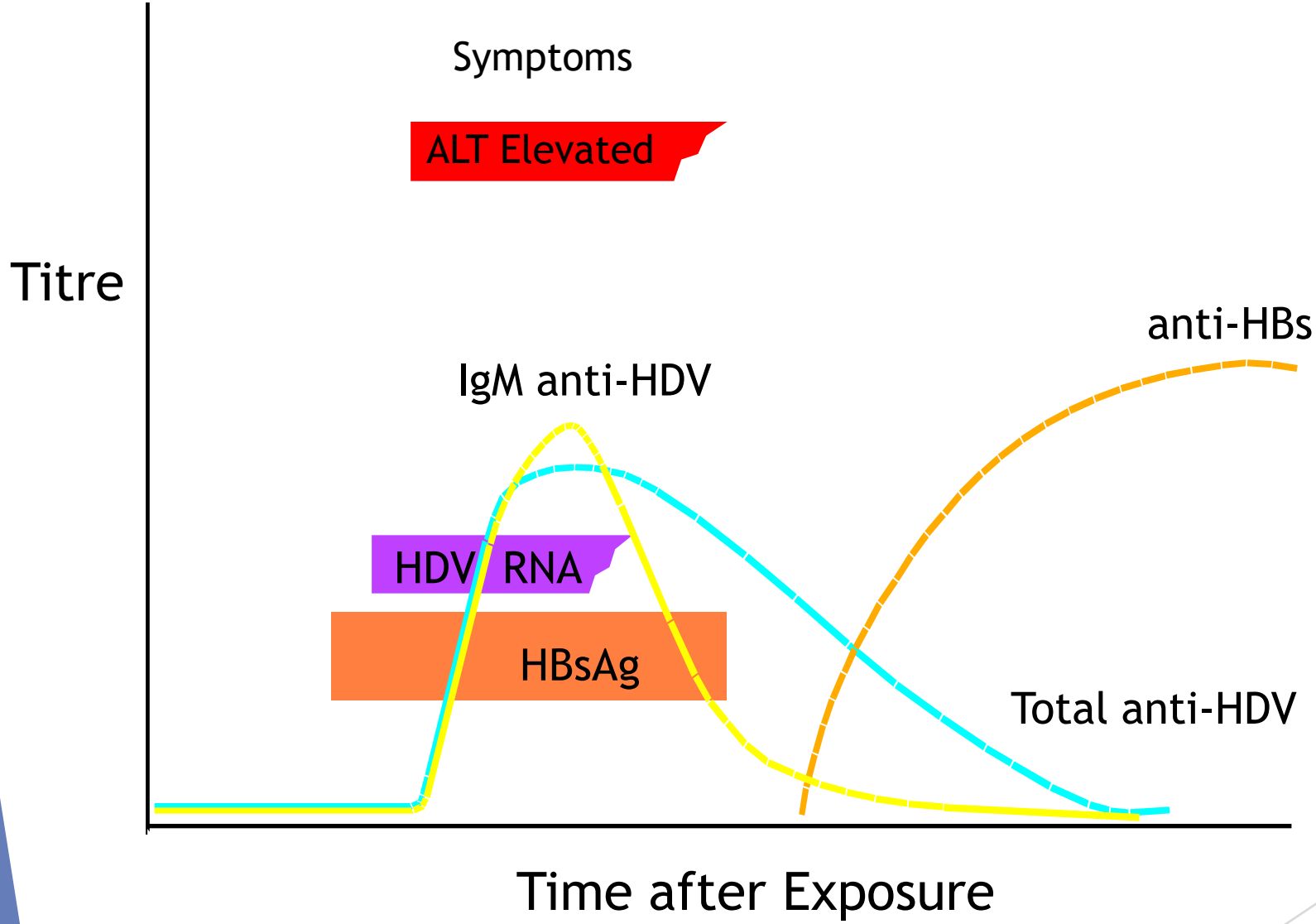
In Superinfection, delta infection occurs in a person already harbouring HBV.

❖Diagnosis

- ✓ Delta antigen is primarily expressed in liver cell nuclei, where it can be demonstrated by immunofluorescence.
- ✓ Anti-delta antibodies appear in serum and can be identified by ELISA.
  IgM antibody appears 2-3 weeks after infection and is soon replaced by the IgG antibody in acute delta infection.

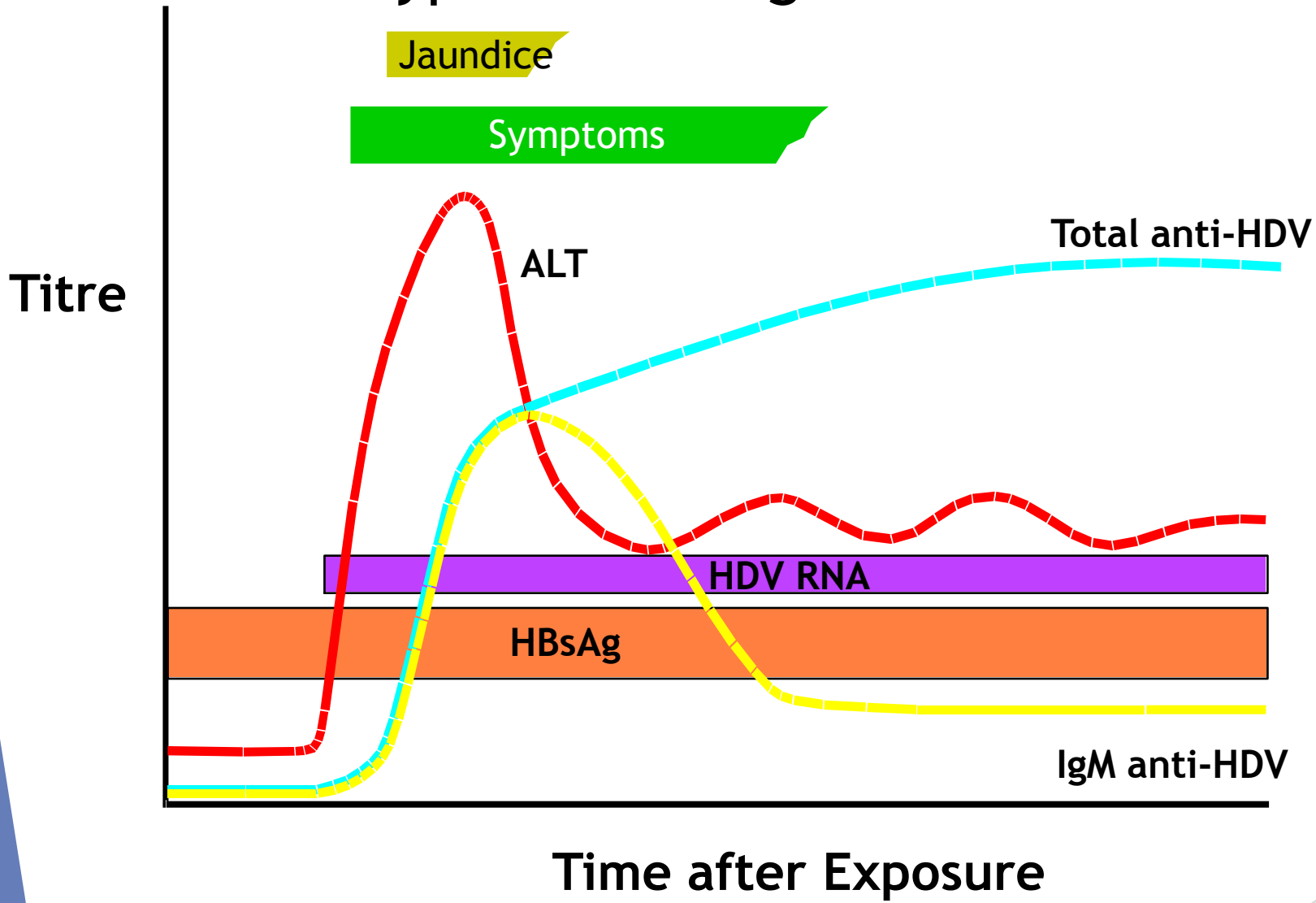
HBV - HDV Coinfection

Typical Serologic Course



HBV - HDV Superinfection

Typical Serologic Course



❖Prevention

- HBV-HDV Coinfection

  Pre or post exposure prophylaxis to prevent HBV infection. Screening of blood donor for HBsAg.

- HBV-HDV Superinfection

  Education to reduce risk behaviors among persons with chronic HBV infection.

Hepatitis E

❖Hepatitis E

- ✓ **Hepatitis E** is a viral hepatitis (liver inflammation) caused by infection with a virus called hepatitis E virus **(HEV).**
- ✓ Although Hepatitis E often causes an acute and self-limiting infection (in that it usually goes away by itself and the patient recovers) with low mortality rates.
- ✓ It bears a high risk of developing chronic hepatitis in immunocompromised patients with substantial mortality rates.
- ✓ Hepatitis E occasionally develops into an acute, severe liver disease, and is fatal in about 2% of all cases.
- ✓ In pregnant women the disease is more often severe and is associated with a clinical syndrome called fulminant hepatic failure.



❖Signs and Symptoms

Acute Infections:

→ The incubation period of hepatitis E varies from 3 to 8 weeks.

→ After a short prodromal phase symptoms lasting from days to weeks follow. They may include jaundice, fatigue and nausea.

→Viral RNA becomes detectable in stool and blood serum during incubation period.

→Serum IgM and IgG antibodies against HEV appear just before onset of clinical symptoms.

→Recovery leads to virus clearance from the blood, while the virus may persist in stool for much longer.

→Recovery is also marked by disappearance of IgM antibodies and increase of levels of IgG antibodies.

Chronic Infections:

→While usually an acute disease, in immunocompromised subjects—particularly in solid organ transplanted patients—hepatitis E may cause a chronic infection.

→Occasionally this may cause liver fibrosis and cirrhosis.

❖Hepatitis E virus

- ✓ HEV is spherical nonenveloped virus, 29-nm to 32 nm in diameter, with a ssRNA genome.
- ✓ The surface of the virion shows indentation and spikes.
- ✓ The Virus is very labile.
- ✓ II has been classified in the genus Herpes virus under the family Caliciviridae.

❖Animal Reservoir: Pigs

❖Incubation Period

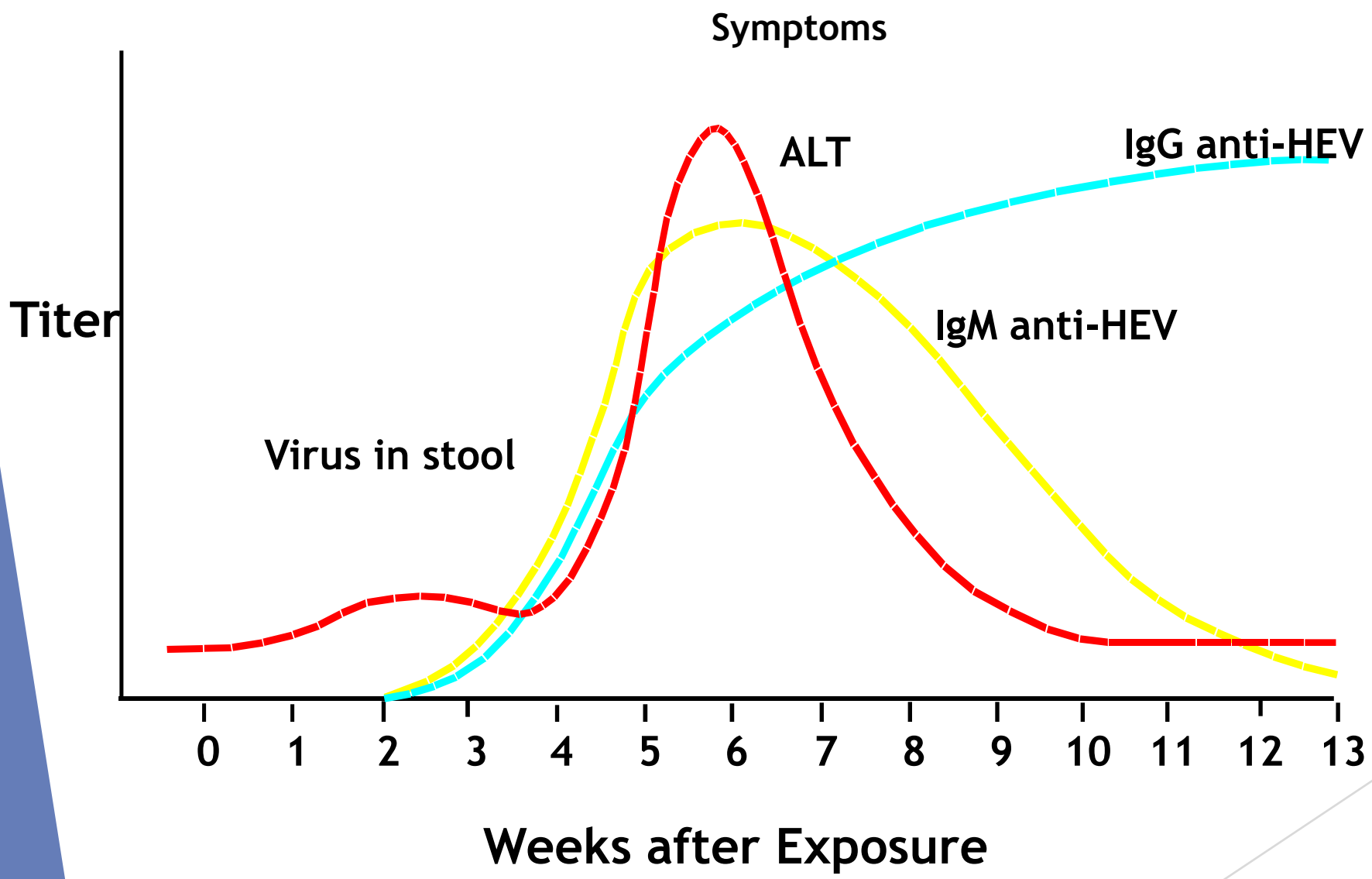
2-9 weeks

❖Mode of Transmission

It is spread mainly by the fecal-oral route due to fecal contamination of water supplies or food; person-to-person transmission is uncommon.

Hepatitis E Virus Infection

Typical Serologic Course



❖Diagnosis

ELISA kits are available for IgG and IgM antibodies, using recombinant and synthetic peptide antigens.

❖Prevention

Sanitation:

Avoid drinking water of unknown purity, uncooked shellfish, and uncooked fruit/vegetables not peeled or prepared by traveler.

❖Hepatitis G

- ✓ **GB virus C** (**GBV-C**), formerly known as **hepatitis G virus** (**HGV**) and also known as HPgV is a virus in the *Flaviviridae* family and a member of the *Pegivirus* genus, is known to infect humans, but is not known to cause human disease.
- ✓ HGV RNA has been found in patients with acute, chronic and fulminant hepatitis, hemophiliacs, patients with multiple transfusions and hemodialysis, intravenous drug addicts and blood donors.



Thank You